Module 1: Network + Network Fundamentals and building networks

Day 5

Task 1:

```
aashanapatel@192 ~ % ping www.youtube.com -c 4
PING www.youtube.com (142.251.156.4): 56 data bytes
64 bytes from 142.251.156.4: icmp_seq=0 ttl=117 time=16.977 ms
64 bytes from 142.251.156.4: icmp_seq=1 ttl=117 time=24.019 ms
64 bytes from 142.251.156.4: icmp_seq=2 ttl=117 time=15.964 ms
64 bytes from 142.251.156.4: icmp_seq=3 ttl=117 time=17.671 ms

--- www.youtube.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 15.964/18.658/24.019/3.154 ms
aashanapatel@192 ~ %
```

Average response time - 18.658 ms

Task 2:

```
aashanapatel@192 ~ % traceroute www.flipkart.com
traceroute: Warning: www.flipkart.com has multiple addresses; using 23.64.140.25
0
traceroute to e348826.dsca.akamaiedge.net (23.64.140.250), 64 hops max, 52 byte
packets
 1  gpon.net (192.168.1.1)  4.136 ms  3.609 ms  3.616 ms
 2  10.130.192.1 (10.130.192.1)  4.580 ms  5.063 ms  4.440 ms
 3  * * *
 4  136.232.112.109 (136.232.112.109)  13.269 ms  16.482 ms  15.291 ms
 5  * * *
 6  * * *
 7  a23-64-140-129.deploy.static.akamaitechnologies.com (23.64.140.129)  15.834
ms  13.635 ms  13.330 ms
 8  a23-64-140-250.deploy.static.akamaitechnologies.com (23.64.140.250)  14.820
ms  13.106 ms  13.441 ms
```

Task 3:

```
aashanapatel@192 ~ % nslookup www.zomato.com
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
www.zomato.com  canonical name = zomato.edgekey.net.
zomato.edgekey.net      canonical name = e70929.dsca.akamaiedge.net.
Name:   e70929.dsca.akamaiedge.net
Address: 49.44.140.187
Name:   e70929.dsca.akamaiedge.net
Address: 49.44.140.145
```

Task 4:

If online: you'll get replies with low response times (usually under 5ms on local Wi-Fi).
If offline: you'll get Request timed out (Windows) or Destination Host Unreachable / 100% packet loss (Mac/Linux) — this means no device is responding at that IP, either because it's powered off, disconnected from Wi-Fi, or blocking ping requests.

Task 5:

For **website not loading**, ping is the fastest first check — for example, pinging youtube.com and getting replies with an average of 18.658 ms confirms basic connectivity is fine, so the issue is likely with the browser or the site itself, not your network.
For **slow connections**, traceroute is more useful — my trace to flipkart.com showed 64 hops, and if one hop shows a much higher delay than the others, that's usually where the slowdown is happening.
For **suspected DNS issues**, nslookup is the right tool — running it on zomato.com returned 192.168.1.53, confirming DNS resolution is working; if it had failed to return an address, that would point to a DNS problem rather than a connectivity one.